Abdurrahman Alyajouri

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SKILLS

- Computer Languages: TypeScript, HTML, CSS, C, C++, Python, SQL.
- Tools: Linux Command Line Interface, Git, Vim, Jira, Visual Studio Code.
- Familiarities: React, Next, Node, Express, Three.js, OpenGL, WebGL, Supabase.

EDUCATION

Portland State University **Expected June 2027**
Bachelor of Science, Computer Science *Portland, Oregon*

- GPA - Major: 4.0, Cumulative: 3.82
- Relevant Courses - Programming Methodologies & Software Implementation, Algorithms & Complexity, Elements of Software Engineering, Intro to Operating Systems, Internetworking Protocols, Intro to Database Management Systems, Principles of Programming Languages, Web & Cloud Security, Physical AI

PROFESSIONAL EXPERIENCE

Software Engineering Capstone (Portland State University) **January 2026 - June 2026**
Software Engineer

- Developed a full-stack web application for a client using TypeScript, Next, React, and Supabase, contributing across frontend, backend, and database layers.
- Designed PostgreSQL schemas and implemented Row Level Security (RLS) policies to ensure secure and controlled data access.
- Collaborated in agile sprints with a team using Jira for task management and weekly sprint planning.

Computer Action Team **October 2025 - Present**
Volunteer Web Developer *Portland, Oregon*

- Developing internally and externally facing web applications and tools for the Maseeh College of Engineering and Computer Science as part of the Computer Action Team, using TypeScript, React, and Node.

LocEssentials **June 2025 - August 2025**
Backend Software Development Intern *Remote*

- Utilized JavaScript and LLM inference via the DeepSeek API to automate a subset of a corpus ingestion pipeline for an automatic translation tool, contributing to reducing manual processing time from hours to minutes, and increasing efficiency for 20+ workshop users.
- Collaborated in agile-style weekly team meetings to coordinate development, address blocking issues, and respond to shifting priorities alongside team members.

PROJECTS

State of the Earth - *TypeScript, React, Node, Express, Three.js, WebGL* **In Progress**

- Designing a near real-time telemetry visualization platform using Server-Sent Events (SSE) and delta updates to efficiently synchronize client state while reducing bandwidth and server load.
- Implementing a WebGL and Three.js rendering layer to support interpolation and visualization of thousands of dynamic geospatial data points in real-time.

Universe Simulator - *C++, OpenGL (Private project, visuals available via LinkedIn)* **In Progress**

- Developed a general purpose C++ shader wrapper for OpenGL 4.3 that compiles all shader types and supports runtime code reloading, reducing iteration time during shader development.
- Implemented a custom GLSL preprocessor enabling file-level “#include” support, further streamlining shader iteration and code reuse.
- Modularized rendering by decoupling the renderer backend from game logic, allowing independent evolution of both systems.

Basic Regex Engine - *C++* **Completed**

- Designed and implemented a memory safe data structure representing non-deterministic finite state automata (NFAs) in order to store the result of a regular expression to NFA conversion operation.
- Developed a suite of overloaded operators to allow ease of use with the NFA data structure.
- Implemented a matching algorithm to permit practical use of the regex engine, by allowing strings to be verified against a given regular expression pattern.